

9.2 3×3 Orthogonal Matrices



Core skills covered in this section:

- ▶ The identity matrix, rotation matrices, reflection matrices, permutation matrices, and their composites are all orthogonal matrices.
- ▶ Expanding the Gram matrix of an orthogonal matrix using the first perspective of matrix multiplication.
- ▶ Projection matrices composing into the identity matrix.
- ▶ Two-step projection: first scalar projection, then vector projection.
- ▶ The orthogonal complement contains all the vectors perpendicular to the original space.

The previous section started with two examples — the identity matrix and the rotation matrix — to explain 2×2 orthogonal matrices from a planar geometric point of view. This section "raises the dimension," explaining 3×3 orthogonal matrices from a three-dimensional geometric point of view.

The 3×3 Identity Matrix

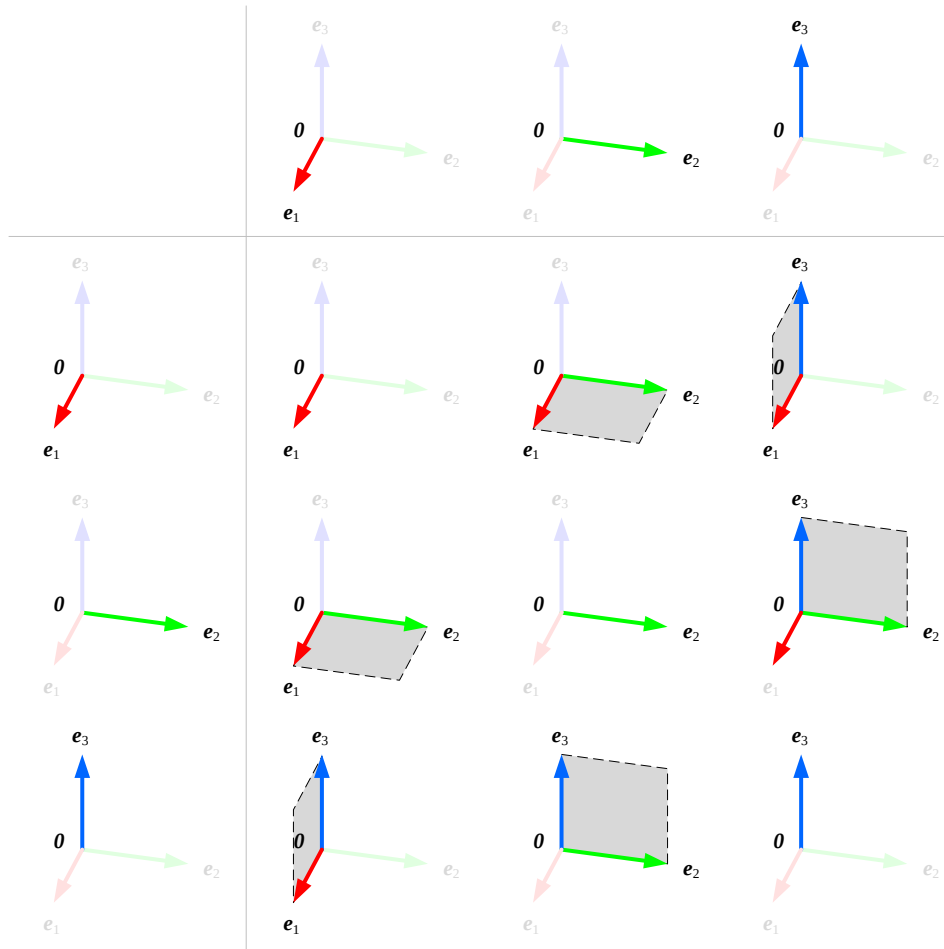
The 3×3 identity matrix I is also an orthogonal matrix, i.e., it satisfies

$$I_{3 \times 3} @ I_{3 \times 3}^T = I_{3 \times 3}^T @ I_{3 \times 3} = I_{3 \times 3}$$

Let's first look at the Gram matrix $I_{3 \times 3}^T @ I_{3 \times 3} = I_{3 \times 3}$, expanding using the first perspective of matrix multiplication

$$\begin{bmatrix} \mathbf{e}_1^T \\ \mathbf{e}_2^T \\ \mathbf{e}_3^T \end{bmatrix} @ [\mathbf{e}_1 \quad \mathbf{e}_2 \quad \mathbf{e}_3] = \begin{bmatrix} \mathbf{e}_1^T @ \mathbf{e}_1 & \mathbf{e}_1^T @ \mathbf{e}_2 & \mathbf{e}_1^T @ \mathbf{e}_3 \\ \mathbf{e}_2^T @ \mathbf{e}_1 & \mathbf{e}_2^T @ \mathbf{e}_2 & \mathbf{e}_2^T @ \mathbf{e}_3 \\ \mathbf{e}_3^T @ \mathbf{e}_1 & \mathbf{e}_3^T @ \mathbf{e}_2 & \mathbf{e}_3^T @ \mathbf{e}_3 \end{bmatrix} = \begin{bmatrix} \mathbf{e}_1 \cdot \mathbf{e}_1 & \mathbf{e}_1 \cdot \mathbf{e}_2 & \mathbf{e}_1 \cdot \mathbf{e}_3 \\ \mathbf{e}_2 \cdot \mathbf{e}_1 & \mathbf{e}_2 \cdot \mathbf{e}_2 & \mathbf{e}_2 \cdot \mathbf{e}_3 \\ \mathbf{e}_3 \cdot \mathbf{e}_1 & \mathbf{e}_3 \cdot \mathbf{e}_2 & \mathbf{e}_3 \cdot \mathbf{e}_3 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I_{3 \times 3}$$

Please use Figure 1 to analyze on your own the characteristics of the main-diagonal and off-diagonal entries in .

Figure 1. Expanding the Matrix Multiplication Using the First Perspective of Matrix Multiplication I/I

Composing the Identity Matrix

Now let's look at $I_{3 \times 3} @ I_{3 \times 3}^T = I_{3 \times 3}$, expanding using the second perspective of matrix multiplication

$$[e_1 \ e_2 \ e_3] @ \begin{bmatrix} e_1^T \\ e_2^T \\ e_3^T \end{bmatrix} = e_1 @ e_1^T + e_2 @ e_2^T + e_3 @ e_3^T = I_{3 \times 3}$$

As Figure 2 shows, what we see is the superposition of three orthogonal projection matrices, and the result is the identity matrix.

For example, $e_1 @ e_1^T @ x$ represents projecting the vector x onto the x_1 axis; $e_2 @ e_2^T @ x$ represents projecting the vector x onto the x_2 axis, and so on. Let's go through each one below.

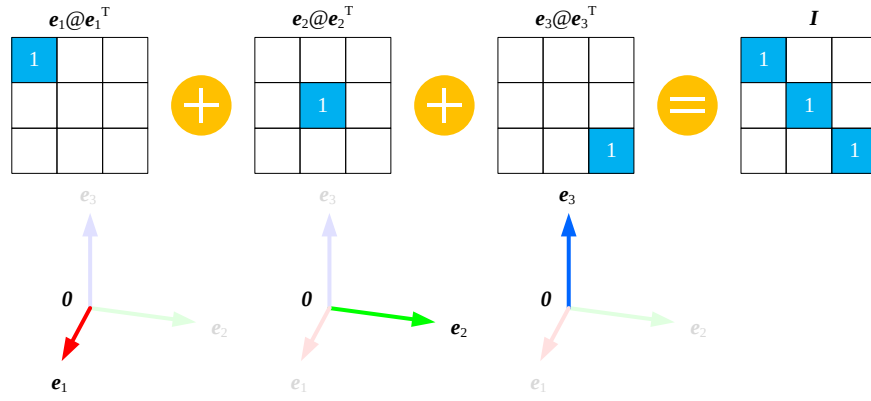


Figure 2. Superimposing Three Orthogonal Projection Matrices Gives the Identity Matrix/

Orthogonal Projection Toward the $\mathbf{e}_1$ Direction

Projecting the three-dimensional column vector $\mathbf{x}$ onto the direction vector (unit vector) $\mathbf{e}_1$, the corresponding projection matrix $\mathbf{P}_1$ is

$$\mathbf{P}_1 = \mathbf{e}_1 @ \mathbf{e}_1^T = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} @ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}^T = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

The projection matrix $\mathbf{P}_1$ is rank one and also a symmetric matrix, i.e., it satisfies

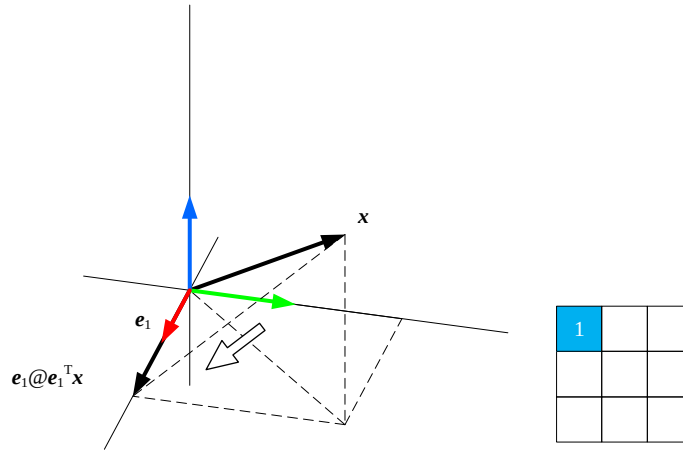
$$(\mathbf{e}_1 @ \mathbf{e}_1^T)^T = \mathbf{e}_1 @ \mathbf{e}_1^T$$

As Figure 3 shows, the projection result corresponds to the following matrix multiplication

$$\mathbf{P}_1 \mathbf{x} = \mathbf{e}_1 @ \mathbf{e}_1^T @ \mathbf{x} = \underbrace{\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}}_{\mathbf{e}_1 @ \mathbf{e}_1^T} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} x_1 \\ 0 \\ 0 \end{bmatrix}$$

The coordinate of the column vector $\mathbf{x}$ in $\text{span}(\mathbf{e}_1)$ is the scalar projection result of $\mathbf{x}$ onto the $\mathbf{e}_1$ direction, i.e.,

$$\mathbf{e}_1^T @ \mathbf{x} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}^T \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = x_1$$

Figure 3. Orthogonal Projection of the Three-Dimensional Column Vector $\mathbf{x}$ Onto the $\mathbf{e}_1$ Direction

Comparing and , we can write the orthogonal projection as two parts

$$\mathbf{P}_1 \mathbf{x} = \underbrace{\mathbf{e}_1}_{\text{Direction}} @ \underbrace{\mathbf{e}_1^T @ \mathbf{x}}_{\text{Scalar}}$$

This is actually a "two-step projection": first project $\mathbf{x}$ onto $\text{span}(\mathbf{e}_1)$, which is the scalar projection; then describe the projection result in three-dimensional space, which corresponds to the vector projection.

Transposing the right-hand side of the equation gives

$$\underbrace{\mathbf{x}^T @ \mathbf{e}_1}_{\text{Scalar}} @ \underbrace{\mathbf{e}_1^T}_{\text{Direction}}$$

The equation above is equivalent to using the row vector $\mathbf{x}^T$ to represent a point in three-dimensional space.

Orthogonal Projection Toward the $\mathbf{e}_2$ Direction

Projecting the three-dimensional column vector $\mathbf{x}$ onto the $\mathbf{e}_2$ direction, the projection matrix $\mathbf{P}_2$ is

$$\mathbf{P}_2 = \mathbf{e}_2 @ \mathbf{e}_2^T = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} @ \begin{bmatrix} 0 & 1 & 0 \end{bmatrix}^T = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

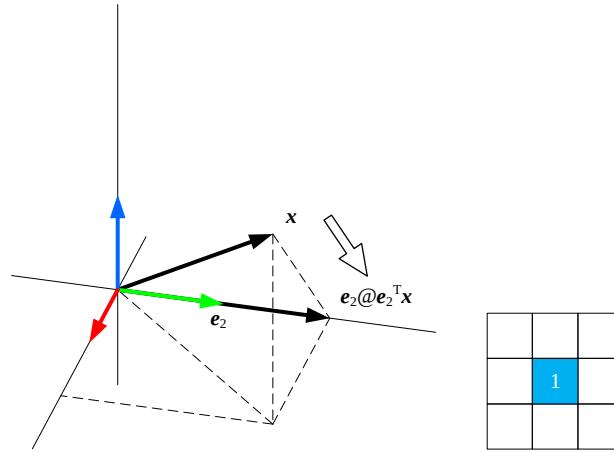
The projection matrix $\mathbf{P}_2$ is likewise rank one and also a symmetric matrix.

As Figure 4 shows, the projection result corresponds to the following matrix multiplication

$$\mathbf{P}_2 \mathbf{x} = \mathbf{e}_2 @ \mathbf{e}_2^T @ \mathbf{x} = \underbrace{\begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}}_{\mathbf{e}_2 @ \mathbf{e}_2^T} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ x_2 \\ 0 \end{bmatrix}$$

The coordinate of the column vector $\mathbf{x}$ in $\text{span}(\mathbf{e}_2)$ is the scalar projection result, i.e.,

$$\mathbf{e}_2^T @ \mathbf{x} = \begin{bmatrix} 0 & 1 & 0 \end{bmatrix}^T \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = x_2$$

Figure 4. Orthogonal Projection of the Three-Dimensional Column Vector $\mathbf{x}$ Onto the $\mathbf{e}_2$ Direction

Orthogonal Projection Toward the $\mathbf{e}_3$ Direction

Projecting the three-dimensional column vector $\mathbf{x}$ onto the $\mathbf{e}_3$ direction, the corresponding projection matrix $\mathbf{P}_3$ is

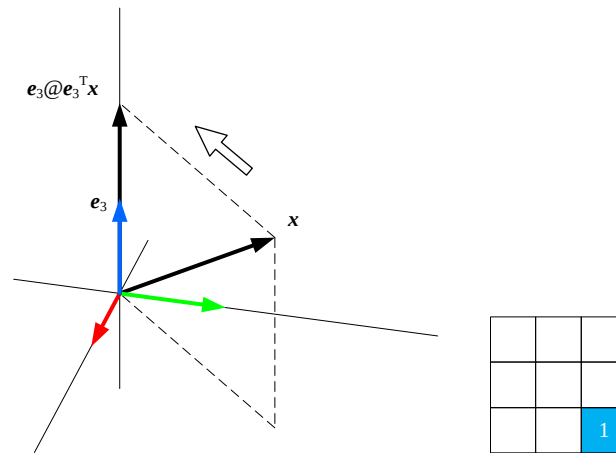
$$\mathbf{P}_3 = \mathbf{e}_3 @ \mathbf{e}_3^T = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} @ \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}^T = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

As Figure 5 shows, the projection result corresponds to the following matrix multiplication

$$\mathbf{P}_3 \mathbf{x} = \mathbf{e}_3 @ \mathbf{e}_3^T @ \mathbf{x} = \underbrace{\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}}_{\mathbf{e}_3 @ \mathbf{e}_3^T} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ x_3 \end{bmatrix}$$

The coordinate of the column vector $\mathbf{x}$ in $\text{span}(\mathbf{e}_3)$ is the scalar projection result, i.e.,

$$\mathbf{e}_3^T @ \mathbf{x} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}^T \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = x_3$$

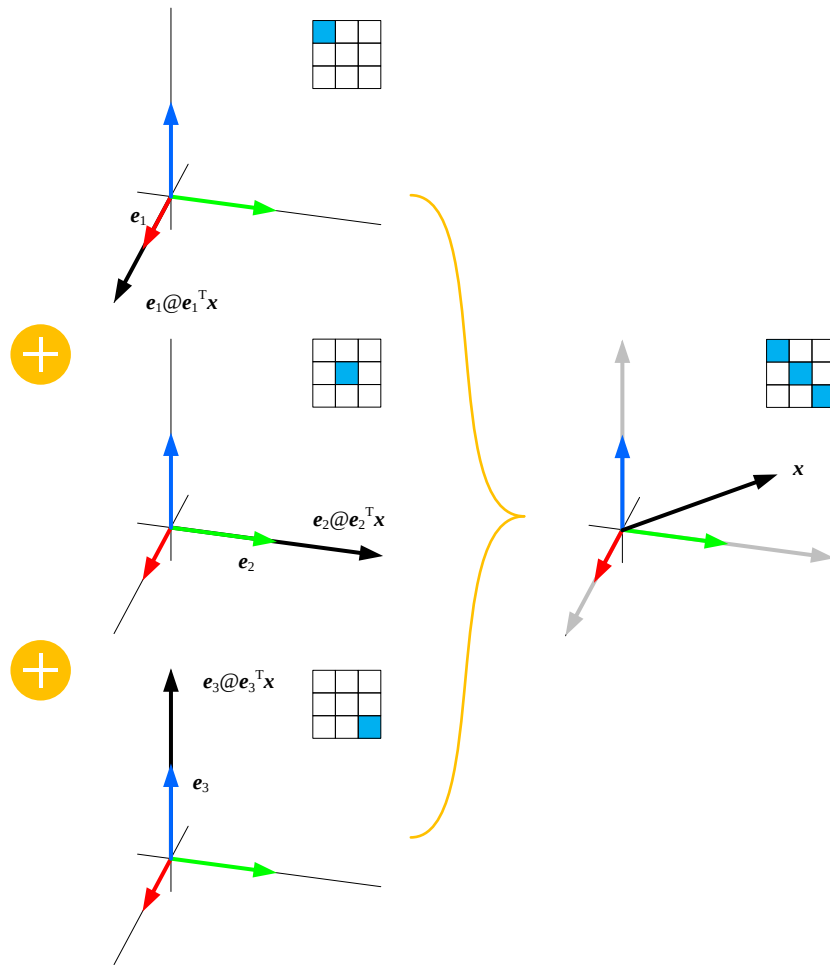
Figure 5. Orthogonal Projection of the Three-Dimensional Column Vector $\mathbf{x}$ Onto the $\mathbf{e}_3$ Direction

Vector Composition

$\mathbf{P}_1\mathbf{x}$, $\mathbf{P}_2\mathbf{x}$, and $\mathbf{P}_3\mathbf{x}$ each contain part of the information in $\mathbf{x}$. However, as Figure 6 shows, superimposing the three orthogonal projection components

$$\mathbf{P}_1\mathbf{x} + \mathbf{P}_2\mathbf{x} + \mathbf{P}_3\mathbf{x} = (\mathbf{P}_1 + \mathbf{P}_2 + \mathbf{P}_3)\mathbf{x} = (\mathbf{e}_1 @ \mathbf{e}_1^T + \mathbf{e}_2 @ \mathbf{e}_2^T + \mathbf{e}_3 @ \mathbf{e}_3^T)\mathbf{x} = \mathbf{I} @ \mathbf{x}$$

we then, within $[\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3]$, recover $\mathbf{x}$ in full!

Figure 6. Recovering the Three-Dimensional Column Vector in $[\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3]\mathbf{x}$

Orthogonal Complement

can be rearranged to give three interesting equations

$$\begin{aligned} \mathbf{I} - \mathbf{e}_1 @ \mathbf{e}_1^T &= \mathbf{e}_2 @ \mathbf{e}_2^T + \mathbf{e}_3 @ \mathbf{e}_3^T \\ \mathbf{I} - \mathbf{e}_2 @ \mathbf{e}_2^T &= \mathbf{e}_1 @ \mathbf{e}_1^T + \mathbf{e}_3 @ \mathbf{e}_3^T \\ \mathbf{I} - \mathbf{e}_3 @ \mathbf{e}_3^T &= \mathbf{e}_1 @ \mathbf{e}_1^T + \mathbf{e}_2 @ \mathbf{e}_2^T \end{aligned}$$

? What do these three matrices represent? Perhaps they remind you of the projection matrices defined using the normal vector.

For example, consider the following equation

$$(\mathbf{I} - \mathbf{e}_1 @ \mathbf{e}_1^T) \mathbf{x} = (\mathbf{e}_2 @ \mathbf{e}_2^T + \mathbf{e}_3 @ \mathbf{e}_3^T) \mathbf{x} = (\mathbf{e}_2 @ \mathbf{e}_2^T) \mathbf{x} + (\mathbf{e}_3 @ \mathbf{e}_3^T) \mathbf{x}$$

The left-hand side represents orthogonal projection onto the x_2x_3 plane.

The x_2x_3 plane's normal vector is $\mathbf{e}_1$, and the corresponding projection matrix for projecting onto this plane is $\mathbf{I} - \mathbf{e}_1 @ \mathbf{e}_1^T$. Specifically, as Figure 7 shows.

The right-hand side of the equation represents projecting separately onto the x_2 and x_3 axes, then superimposing the results. It's worth noting that $\text{span}(\mathbf{e}_1)$ and $\text{span}(\mathbf{e}_2, \mathbf{e}_3)$ are orthogonal complements of each other.

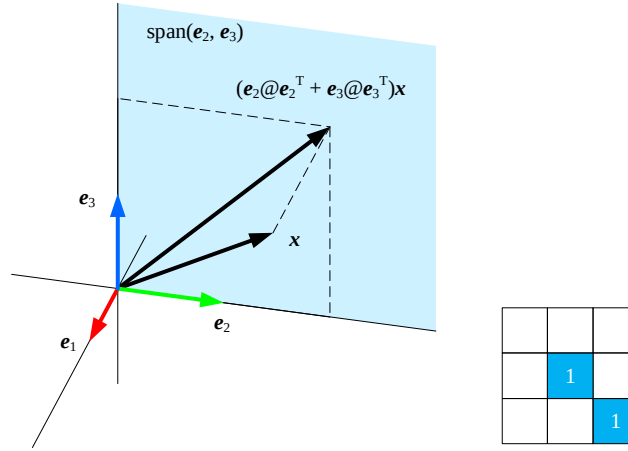


Figure 7. Projecting the Three-Dimensional Column Vector Onto the x_2x_3 Plane

Let's look at another example; consider the following equation

$$(I - \mathbf{e}_2 @ \mathbf{e}_2^T) \mathbf{x} = (\mathbf{e}_1 @ \mathbf{e}_1^T + \mathbf{e}_3 @ \mathbf{e}_3^T) \mathbf{x} = (\mathbf{e}_1 @ \mathbf{e}_1^T) \mathbf{x} + (\mathbf{e}_3 @ \mathbf{e}_3^T) \mathbf{x}$$

The left-hand side represents orthogonal projection onto the x_1x_3 plane, where $\mathbf{e}_2$ is the x_1x_3 plane's normal vector. Figure 8 shows the projection of the three-dimensional column vector onto the x_1x_3 plane.

The right-hand side of the equation represents projecting separately onto the x_1 and x_3 axes, then superimposing the results.

That is, $\text{span}(\mathbf{e}_2)$ and $\text{span}(\mathbf{e}_1, \mathbf{e}_3)$ are orthogonal complements of each other. Orthogonally projecting onto each of the two spaces gives only partial information; combining the two together gives the complete information.

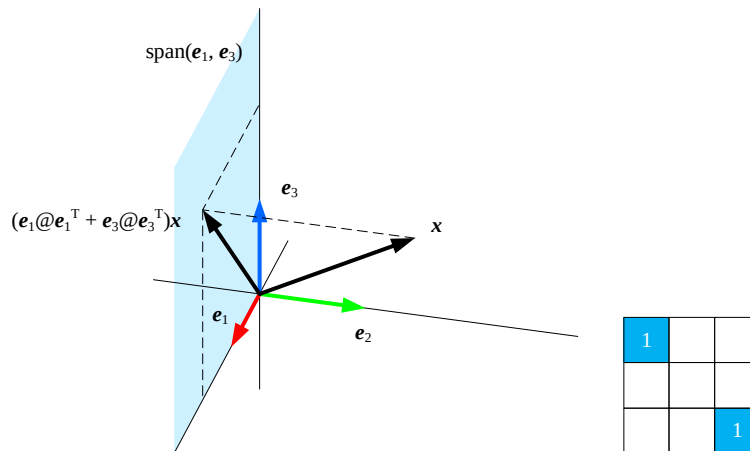


Figure 8. Projecting the Three-Dimensional Column Vector Onto the x_1x_3 Plane

Please analyze on your own the following equation and Figure 9

$$(I - e_3 @ e_3^T) x = (e_1 @ e_1^T + e_2 @ e_2^T) x = (e_1 @ e_1^T) x + (e_2 @ e_2^T) x$$

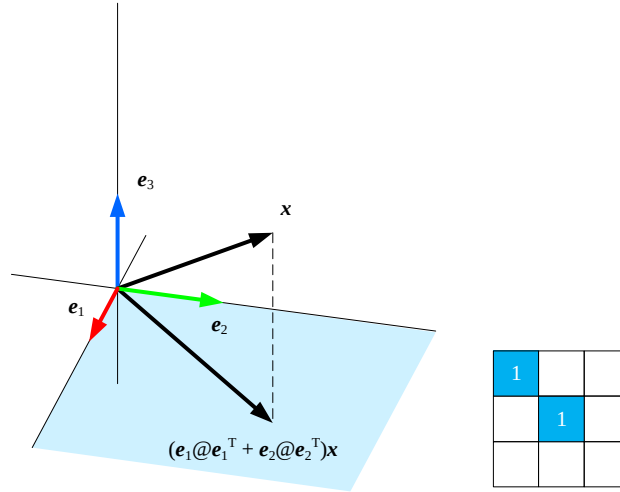


Figure 9. Projecting the Three-Dimensional Column Vector Onto the x_1x_2 Plane

3 × 3 Rotation Matrices

Given the following three-dimensional (about the origin) rotation matrix V

$$V = \begin{bmatrix} 4/5 & -3/5 & 0 \\ 3/5 & 4/5 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 4/5 \\ 3/5 \\ 0 \end{bmatrix}_{v_1} \underbrace{\begin{bmatrix} -3/5 \\ 4/5 \\ 0 \end{bmatrix}}_{v_2} \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}_{v_3}$$

The matrix V is also an orthogonal matrix, satisfying

$$VV^T = V^TV = I$$

Let's first look at the Gram matrix V^TV , writing V as $[v_1, v_2, v_3]$, and expanding using the first perspective of matrix multiplication

$$\begin{bmatrix} v_1^T \\ v_2^T \\ v_3^T \end{bmatrix} @ [v_1 \ v_2 \ v_3] = \begin{bmatrix} v_1^T @ v_1 & v_1^T @ v_2 & v_1^T @ v_3 \\ v_2^T @ v_1 & v_2^T @ v_2 & v_2^T @ v_3 \\ v_3^T @ v_1 & v_3^T @ v_2 & v_3^T @ v_3 \end{bmatrix} = I_{3 \times 3}$$

V^TV 's main-diagonal entries are the inner products of V 's column vectors with themselves, i.e., the squared L^2 norm; V^TV 's off-diagonal entries are V 's pairwise column-vector inner products, all equal to 0, because V 's column vectors are pairwise orthogonal.

Composing the Identity Matrix

Now let's look at VV^T , expanding using the second perspective of matrix multiplication

$$[\mathbf{v}_1 \quad \mathbf{v}_2 \quad \mathbf{v}_3] @ \begin{bmatrix} \mathbf{v}_1^T \\ \mathbf{v}_2^T \\ \mathbf{v}_3^T \end{bmatrix} = \mathbf{v}_1 @ \mathbf{v}_1^T + \mathbf{v}_2 @ \mathbf{v}_2^T + \mathbf{v}_3 @ \mathbf{v}_3^T = \mathbf{I}_{3 \times 3}$$

What we see is the superposition of three orthogonal projection matrices, and the result is the identity matrix.

Orthogonal Projection Toward the $\mathbf{v}_1$ Direction

Projecting the three-dimensional column vector $\mathbf{x}$ onto the direction vector (unit vector) $\mathbf{v}_1$, the corresponding projection matrix $\mathbf{P}_1$ is

$$\mathbf{P}_1 = \mathbf{v}_1 @ \mathbf{v}_1^T = \begin{bmatrix} 4/5 \\ 3/5 \\ 0 \end{bmatrix} @ \begin{bmatrix} 4/5 \\ 3/5 \\ 0 \end{bmatrix}^T = \begin{bmatrix} 16/25 & 12/25 & 0 \\ 12/25 & 9/25 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

The projection matrix above is rank one and also a symmetric matrix.

The projection result corresponds to the following matrix multiplication

$$\mathbf{P}_1 \mathbf{x} = \underbrace{\mathbf{v}_1 @ \mathbf{v}_1^T}_{\mathbf{v}_1 @ \mathbf{v}_1^T} \mathbf{x} = \begin{bmatrix} 16/25 & 12/25 & 0 \\ 12/25 & 9/25 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 16x_1/25 + 12x_2/25 \\ 12x_1/25 + 9x_2/25 \\ 0 \end{bmatrix}$$

The coordinate of the column vector $\mathbf{x}$ in $\text{span}(\mathbf{v}_1)$ is the scalar projection

$$\mathbf{v}_1^T @ \mathbf{x} = \begin{bmatrix} 4/5 \\ 3/5 \\ 0 \end{bmatrix}^T \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 4x_1/5 + 3x_2/5$$

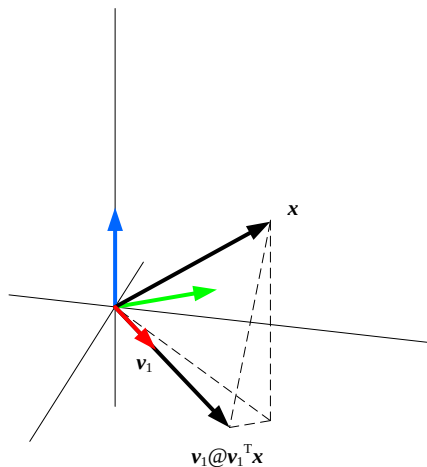


Figure 10. Orthogonal Projection of the Three-Dimensional Column Vector $\mathbf{x}$ Onto the $\mathbf{v}_1$ Direction

Orthogonal Projection Toward the $\mathbf{v}_2$ Direction

Projecting the three-dimensional column vector $\mathbf{x}$ onto the direction vector (unit vector) $\mathbf{v}_2$, the corresponding projection matrix $\mathbf{P}_2$ is

$$\mathbf{P}_2 = \mathbf{v}_2 @ \mathbf{v}_2^T = \begin{bmatrix} -3/5 \\ 4/5 \\ 0 \end{bmatrix} @ \begin{bmatrix} -3/5 \\ 4/5 \\ 0 \end{bmatrix}^T = \begin{bmatrix} 9/25 & -12/25 & 0 \\ -12/25 & 16/25 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

The projection result corresponds to the following matrix multiplication

$$\mathbf{P}_2 \mathbf{x} = \mathbf{v}_2 @ \mathbf{v}_2^T \mathbf{x} = \underbrace{\begin{bmatrix} 9/25 & -12/25 & 0 \\ -12/25 & 16/25 & 0 \\ 0 & 0 & 0 \end{bmatrix}}_{\mathbf{v}_2 @ \mathbf{v}_2^T} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 9x_1/25 - 12x_2/25 \\ -12x_1/25 + 16x_2/25 \\ 0 \end{bmatrix}$$

The coordinate of the column vector $\mathbf{x}$ in $\text{span}(\mathbf{v}_2)$ is the scalar projection result, i.e.,

$$\mathbf{v}_2^T @ \mathbf{x} = \begin{bmatrix} -3/5 \\ 4/5 \\ 0 \end{bmatrix}^T \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = -3x_1/5 + 4x_2/5$$

Orthogonal Projection Toward the $\mathbf{v}_3$ Direction

Projecting the three-dimensional column vector $\mathbf{x}$ onto the direction vector (unit vector) $\mathbf{v}_3$, the corresponding projection matrix $\mathbf{P}_3$ is

$$\mathbf{P}_3 = \mathbf{v}_3 @ \mathbf{v}_3^T = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} @ \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}^T = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

The projection result corresponds to the following matrix multiplication

$$\mathbf{P}_3 \mathbf{x} = \mathbf{v}_3 @ \mathbf{v}_3^T \mathbf{x} = \underbrace{\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}}_{\mathbf{v}_3 @ \mathbf{v}_3^T} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ x_3 \end{bmatrix}$$

The coordinate of the column vector $\mathbf{x}$ in $\text{span}(\mathbf{v}_3)$ is the scalar projection result, i.e.,

$$\mathbf{v}_3^T @ \mathbf{x} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}^T \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = x_3$$

Vector Composition

Superimposing $\mathbf{v}_1$, $\mathbf{v}_2$, and $\mathbf{v}_3$ recovers $\mathbf{x}$ in full

$$(\mathbf{v}_1 @ \mathbf{v}_1^T + \mathbf{v}_2 @ \mathbf{v}_2^T + \mathbf{v}_3 @ \mathbf{v}_3^T) \mathbf{x} = \begin{bmatrix} 16x_1/25 + 12x_2/25 \\ 12x_1/25 + 9x_2/25 \\ 0 \end{bmatrix} + \begin{bmatrix} 9x_1/25 - 12x_2/25 \\ -12x_1/25 + 16x_2/25 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ x_3 \end{bmatrix} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

Orthogonal Complement

can also be rearranged to give three interesting equations

$$\begin{aligned} I - \mathbf{v}_1 @ \mathbf{v}_1^T &= \mathbf{v}_2 @ \mathbf{v}_2^T + \mathbf{v}_3 @ \mathbf{v}_3^T \\ I - \mathbf{v}_2 @ \mathbf{v}_2^T &= \mathbf{v}_1 @ \mathbf{v}_1^T + \mathbf{v}_3 @ \mathbf{v}_3^T \\ I - \mathbf{v}_3 @ \mathbf{v}_3^T &= \mathbf{v}_1 @ \mathbf{v}_1^T + \mathbf{v}_2 @ \mathbf{v}_2^T \end{aligned}$$

Let's first look at the first equation,

$$(I - \mathbf{v}_1 @ \mathbf{v}_1^T) \mathbf{x} = (\mathbf{v}_2 @ \mathbf{v}_2^T) \mathbf{x} + (\mathbf{v}_3 @ \mathbf{v}_3^T) \mathbf{x}$$

The left-hand side represents orthogonal projection onto the $\text{span}(\mathbf{v}_2, \mathbf{v}_3)$ plane, whose normal vector is $\mathbf{v}_1$.

The right-hand side of the equation represents projecting separately onto $\text{span}(\mathbf{v}_2)$ and $\text{span}(\mathbf{v}_3)$, then superimposing the results. $\text{span}(\mathbf{v}_1)$ and $\text{span}(\mathbf{v}_2, \mathbf{v}_3)$ are orthogonal complements of each other.

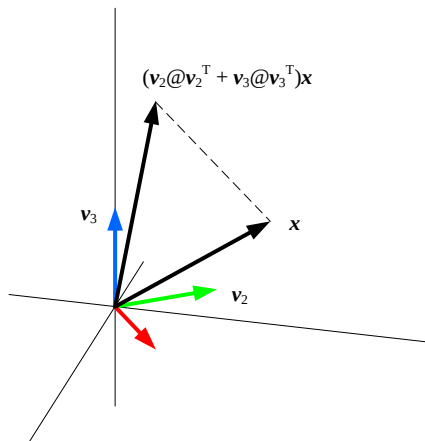


Figure 11. Projecting the Three-Dimensional Column Vector $\mathbf{x}$ Onto $\text{span}(\mathbf{v}_2, \mathbf{v}_3)$

Now let's look at the second equation

$$(I - \mathbf{v}_2 @ \mathbf{v}_2^T) \mathbf{x} = (\mathbf{v}_1 @ \mathbf{v}_1^T) \mathbf{x} + (\mathbf{v}_3 @ \mathbf{v}_3^T) \mathbf{x}$$

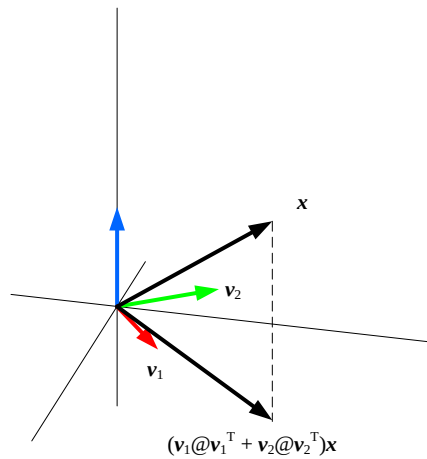
In equation , the left-hand side represents orthogonal projection onto $\text{span}(\mathbf{v}_1, \mathbf{v}_3)$, whose normal vector is $\mathbf{v}_2$.

In equation , the right-hand side represents projecting separately onto $\text{span}(\mathbf{v}_1)$ and $\text{span}(\mathbf{v}_3)$, then superimposing the results. $\text{span}(\mathbf{v}_2)$ and $\text{span}(\mathbf{v}_1, \mathbf{v}_3)$ are orthogonal complements of each other.

Finally, let's look at the third equation

$$(I - \mathbf{v}_3 @ \mathbf{v}_3^T) \mathbf{x} = (\mathbf{v}_1 @ \mathbf{v}_1^T) \mathbf{x} + (\mathbf{v}_2 @ \mathbf{v}_2^T) \mathbf{x}$$

Please analyze on your own the equation and Figure 12.

Figure 12. Projecting the Three-Dimensional Column Vector x Onto $\text{span}(v_1, v_2)$ 

Q1. Please determine which of the following matrices are orthogonal matrices.

▶ $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

▶ $\begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}$

▶ $\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

▶ $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$

▶ $\begin{bmatrix} 1 & 0 \\ 0 & -1 \\ 0 & 0 \end{bmatrix}$

▶ $\begin{bmatrix} 1 & 0 \\ 0 & 0 \\ 0 & 1 \end{bmatrix}$

▶ $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

▶
$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$$

▶
$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

▶
$$\begin{bmatrix} \sqrt{2}/2 & -\sqrt{2}/2 & 0 \\ \sqrt{2}/2 & \sqrt{2}/2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Q2. From a geometric point of view, what geometric transformations do the orthogonal matrices in Q1 correspond to?